A Data-Driven Investment Framework for Enhancing Preventive Health in the UAE

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Executive Summary

The United Arab Emirates (UAE) is at a pivotal moment of opportunity, poised to build upon its remarkable healthcare achievements by addressing the evolving challenge of chronic diseases. These non-communicable diseases (NCDs) represent a significant health and economic consideration for the nation's future. The current healthcare model, with a primary focus on curative treatment, presents an opportunity for greater alignment with the country's forward-looking, prevention-oriented strategic goals.

This report highlights an opportunity to further align resource allocation with the nation's strategic vision. An analysis of current health expenditure reveals that while national strategies

like "We the UAE 2031" champion prevention, approximately 1% of health expenditure is allocated to it, compared to 57% for curative care (1). This pattern often arises when decision-making systems lack a standardized framework to quantitatively demonstrate the long-term value of near-term investments in prevention. To bridge this gap, this report presents a comprehensive economic evaluation demonstrating that a scaled national investment in preventive medicine is not a cost, but one of the highest-value capital investments the nation can make.

The analysis reveals that such an investment could yield a 157% return on investment (ROI) over 10 years. A total investment of AED 20.4 billion is projected to generate AED 52.4 billion in direct healthcare savings and societal benefits, resulting in a net societal benefit of AED 32.0 billion.

The population health impact of this strategic shift would be transformative. The proposed portfolio of interventions would prevent an estimated 158,080 major disease events—including heart attacks, strokes, and cancer diagnoses—and avert 16,325 premature deaths over a decade. Furthermore, these programs would generate 326,280 Quality-Adjusted Life Years (QALYs), representing hundreds of thousands of years of life lived in better health by the people of the UAE. All modeled interventions prove to be highly cost-effective, with an aggregate cost per QALY of AED 62,600, a figure well below the UAE's established willingness-to-pay threshold of approximately AED 150,000 (2, 3, 4).

As the central strategic enabler for this transformation, this report proposes the development and implementation of the Preventive Medicine ROI Calculator. This evidence-based analytical tool is designed specifically for the UAE context to guide policy, justify budget reallocations, and align public and private sector efforts. The calculator will provide a standardized framework to quantify the ROI of preventive health interventions, directly supporting the implementation of the UAE's new Health Technology Assessment (HTA) guidelines and cementing data-driven governance at the heart of the health system (5, 6).

The evidence presented is clear. The following key recommendations provide a path forward:

- Authorize and fund the development and phased implementation of the Preventive Medicine ROI Calculator as a national strategic priority.
- Consider a National Prevention Act to establish universal entitlements to preventive services and create sustainable, multi-source funding structures.
- Commit to a phased budgetary allocation for high-ROI preventive interventions, reaching AED 4.5 billion annually by the fifth year of implementation.

- Launch a dedicated public-private partnership initiative, leveraging the ROI Calculator to engage the business community as a key partner in funding and delivering preventive health solutions.

By adopting this data-driven framework, the UAE can advance its transition from a reactive treatment model to a proactive, high-value prevention paradigm, securing not only the long-term fiscal sustainability of its healthcare system but also the health, well-being, and prosperity of the nation for generations to come.

Section 1: The National Opportunity: Enhancing the UAE's Future Health and Prosperity

The United Arab Emirates has established itself as a global leader in economic development and innovation, a success underpinned by a deep commitment to the well-being of its population. This commitment is reflected in a world-class healthcare system that delivers high standards of clinical care. The very socioeconomic transformations that have driven the nation's progress have also given rise to an evolving public health challenge: a growing prevalence of chronic, non-communicable diseases. This section establishes the scale of this challenge, its economic dimensions, and the compelling opportunity for a strategic, data-driven focus on preventive medicine, in alignment with the nation's highest strategic goals.

1.1 Building on the National Vision

The leadership of the UAE has clearly articulated a future where public health is defined not by the treatment of illness, but by its prevention. This forward-looking perspective is embedded in the nation's core strategic documents. The recommendations detailed in this report are directly aligned with and in service of the highest national strategies, including "We the UAE 2031" and the UAE Centennial 2071, both of which envision a society that enjoys a high quality of life, with health and well-being as central pillars of a competitive, knowledge-based economy (7, 8).

Realizing this ambition involves a continued shift from a reactive healthcare model to one that is proactive and prevention-led. The Ministry of Health and Prevention (MOHAP) has translated this vision into a clear operational mandate. Its mission to "Govern an integrated preventive and therapeutic healthcare system" is supported by a strategic objective for 2023-2026 to "upgrade the level of public health for a preventive, aware, safe and committed community" (9). This focus is further solidified by specific national policies, most notably the National Policy for

Promoting Healthy Lifestyles, issued in 2022. This policy directly targets the primary drivers of NCDs—unhealthy eating, physical inactivity, and tobacco use—and sets tangible goals for risk reduction (10).

While initiatives like the Abu Dhabi Public Health Centre's "IFHAS" program for early disease detection are tangible steps, fully realizing this vision requires that strategic intent be supported by a robust mechanism for resource allocation (11). This report provides the evidence-based, quantitative justification needed to translate the UAE's prevention-first vision into a funded and operational reality.

1.2 Addressing the Evolving Challenge of Non-Communicable Diseases (NCDs)

The strategic imperative for prevention is informed by epidemiological trends. The UAE has undergone a rapid epidemiological transition, shifting from communicable to non-communicable diseases as the primary health consideration within just five decades. These NCDs now account for between 55% and 68% of all deaths in the country (12, 13, 14).

- **Cardiovascular Disease (CVD):** As the leading cause of NCD-related mortality, CVD is responsible for 34% of such deaths and over 25% of all deaths in the nation (12, 15, 16, 17). A critical factor is the premature onset of CVD in the population. Local medical experts have noted that the first major cardiac event, such as a heart attack, occurs in the UAE population 10 to 15 years earlier than the international average, indicating that the most productive years of the population are being significantly impacted (18, 19). This is driven by a high prevalence of risk factors, with studies documenting hypertension in 22.4% of young adults, dyslipidemia in 62.7%, and central obesity in 22.5% (20).
- **Diabetes:** The UAE has one of the highest rates of type 2 diabetes in the world. The International Diabetes Federation reports a prevalence of 12.3% among adults, translating to approximately 990,000 cases in 2021, with other sources citing an age-adjusted prevalence as high as 20.7% (12, 15, 21, 22). Projections suggest that diabetes cases could double by 2040 and prevalence could climb to 21.4% by 2030 (15, 23). Compounding this issue is a significant rate of undiagnosed cases; an estimated 35% to 64% of individuals with diabetes are unaware of their condition, forgoing early management and progressing toward more complex health stages (24, 25).
- **Obesity:** The challenges of diabetes and CVD are largely fueled by a foundational prevalence of obesity. The UAE faces high rates of overweight and obesity, with a

quarter of all adults currently living with obesity and nearly 40% of children being either overweight or obese (26, 27, 28). Projections from the World Obesity Federation estimate that by 2035, nearly 7.5 million people in the UAE will be classified as overweight or obese. This trend is a direct contributor to mortality, with 7,622 deaths in 2019 linked to excess weight (26).

The premature onset of these diseases creates a unique economic dynamic. In many nations, chronic diseases manifest later in life, often post-retirement, where the primary cost is direct healthcare expenditure. In the UAE, a preventable heart attack at an early age strikes an individual at the peak of their professional career and economic contribution. This means the indirect cost of a single disease event, measured in lost productivity, is significantly higher and is realized much faster. This dynamic amplifies the "Return" side of the ROI equation, transforming prevention from a long-term societal good into a near-to-medium-term economic consideration for national prosperity and competitiveness.

1.3 The Economic Case for Proactive Health Investment

The human dimension of the NCD challenge is mirrored by a significant economic one. The total economic impact of NCDs, including healthcare expenditures and productivity losses, is estimated at approximately AED 39.9 billion (US$10.9 billion) per year, equivalent to 2.7% of the country's 2019 GDP (12, 29). Cardiovascular disease alone produces economic effects of approximately AED 26 billion, representing two-thirds of the total NCD impact (12).

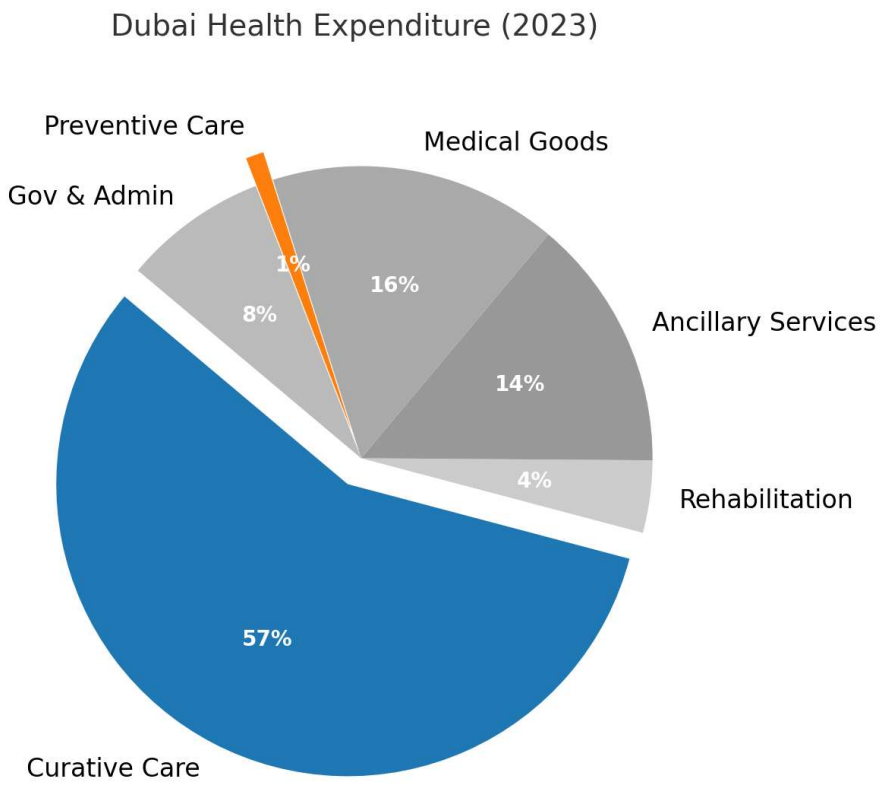
The financial trajectory of specific diseases illustrates the logic for proactive investment:

- **Diabetes Costs:** In 2021, the total healthcare cost for diabetes in the UAE was $2.09 billion (AED 7.68 billion). Modeling projects this annual figure could grow to $3.4 billion (AED 12.49 billion) by 2031 (23, 30). The cost per patient escalates dramatically with the onset of complications, rising from AED 9,200 annually for uncomplicated cases to AED 55,334 for patients with multiple complications (31).
- **Obesity Costs:** The costs associated with overweight and obesity are even more expansive. The World Obesity Federation projects that the total economic impact of obesity in the UAE will reach nearly 5% of the nation's GDP by 2035, with annual costs approaching $12 billion (26). This figure encompasses not only direct healthcare expenditure but also significant indirect costs from lost productivity, which are projected to climb to $30.7 billion by 2035 if current trends continue (32).

1.4 Optimizing Resource Allocation to Align with Strategic Vision

An examination of the UAE's health financing reveals an opportunity for optimization. In Dubai, for example, total health expenditure reached AED 22.24 billion in 2023 (1). Analysis of these allocations shows that while national strategies champion prevention, the financial flows reflect a different emphasis. Approximately 57% of total health expenditure (AED 12.68 billion) is directed toward curative care—treating diseases that are already established. In contrast, 1% (AED 167 million) is allocated to preventive care (1).

This allocation pattern is not a reflection of strategic intent but is often a symptom of an information gap. Policymakers and budget holders, faced with the immediate and tangible costs of treating sick patients, may lack a standardized, trusted, and quantitative framework to justify reallocating significant funds toward preventive measures whose financial benefits are realized over the long term. This can create a cycle where the underfunding of prevention leads to a steady pipeline of new chronic disease patients, which in turn consumes a larger share of the budget for curative care. The analysis and tools proposed in this report are designed specifically to help break this cycle by making the future financial case for prevention tangible, defensible, and actionable today.



Category	Percentage of Total Expenditure	Amount (AED billion)
Curative Care	57%	12.68
Rehabilitation	4%	0.89
Ancillary Services	14%	3.11
Medical Goods	16%	3.56
Preventive Care	1%	0.17
Governance & Administration	8%	1.78

Source: Data derived from Dubai Health Authority (1). Note: Percentages may not sum to 100% due to rounding.		

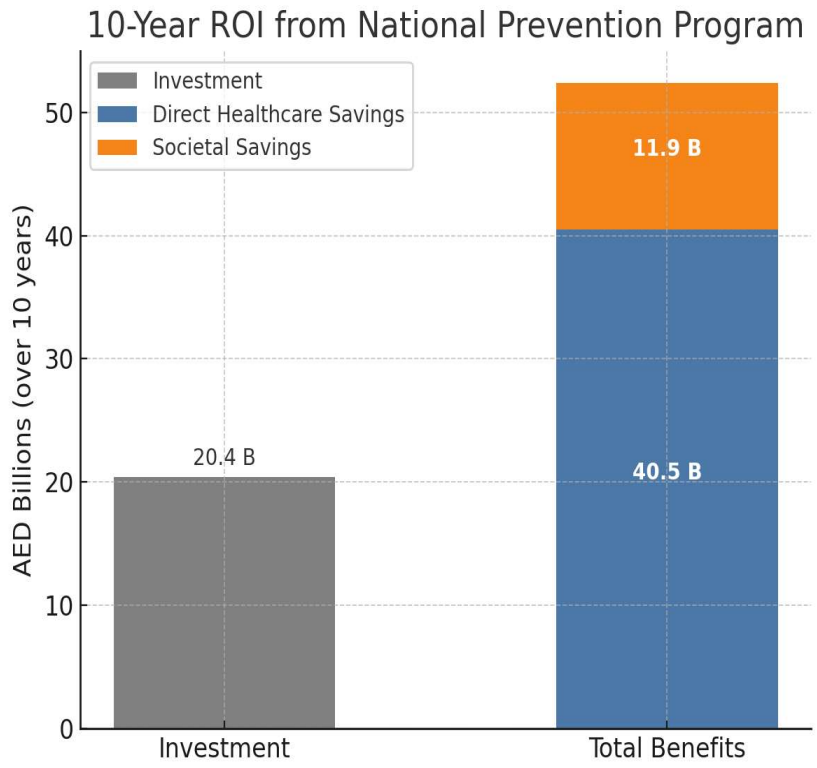
Section 2: Quantifying the Opportunity: A Prevention-Led Paradigm

The scale of the NCD challenge and its associated economic dimensions underscore the need for a continued evolution in how the UAE's healthcare system allocates its resources. The findings establish the immense scale of the opportunity, demonstrating that a strategic investment in prevention can yield transformative returns for the nation's health and its economy.

2.1 Aggregate Health and Economic Outcomes

A comprehensive, scaled investment in preventive programs across five key disease areas—cardiovascular disease, type 2 diabetes, cancer, Alzheimer's disease, and osteoporosis—can generate exceptional returns. The economic modeling demonstrates a 157% Return on Investment (ROI) over a 10-year horizon. This is calculated as the net societal benefit (Total Benefits minus Total Investment) divided by the Total Investment (AED 32.0 billion/AED 20.4 billion).

This powerful financial return is the result of a carefully modeled comparison between investment and savings. A total investment of AED 20.4 billion over 10 years is projected to generate AED 40.5 billion in direct healthcare savings by averting costly treatments, hospitalizations, and long-term care. In addition, it is projected to generate AED 11.9 billion in societal savings, primarily through preserved productivity from a healthier workforce. The total combined benefit of AED 52.4 billion far outweighs the initial investment, resulting in a net societal benefit of AED 32.0 billion for the UAE.



Economic Metric	Projected Value (Discounted)	95% Confidence Interval
Total Investment	AED 20.4 billion	N/A (Fixed Budget)
Total Benefits	AED 52.4 billion	AED 44.8 billion - AED 60.0 billion
Net Societal Benefit	AED 32.0 billion	AED 19.9 billion - AED 45.9 billion
Return on Investment (ROI)	157%	119% - 196%
Source: Data derived from comprehensive Markov modeling analysis conducted for this report.		

2.2 A Transformation in Population Health

Beyond the compelling financial returns, the human impact of this investment is profound. The implementation of these preventive programs is projected to fundamentally alter the trajectory of chronic disease in the UAE, leading to significant gains in both length and quality of life for the population.

Over the 10-year analysis period, the programs are projected to:

- **Prevent 158,080 major disease events.** This includes averting thousands of life-altering events such as heart attacks, strokes, new cancer diagnoses, and debilitating osteoporotic fractures.
- **Prevent 16,325 premature deaths.** This represents a profound societal value that extends far beyond any economic calculation, preserving families and communities.
- **Generate 326,280 Quality-Adjusted Life Years (QALYs).** This is the gold-standard metric for health gain, representing more than three hundred thousand years of life lived in better health, free from the disability and suffering of chronic disease.

Health Impact Metric	Projected Outcome	95% Confidence Interval
Major Disease Events Prevented	158,080	142,000 - 174,000
Premature Deaths Prevented	16,325	14,000 – 18,500
Quality-Adjusted Life Years (QALYs) Gained	326,280	286,000 - 367,000
Source: Data derived from comprehensive Markov modeling analysis conducted for this report.		

2.3 Demonstrating High Value through Cost-Effectiveness

A critical measure of value in healthcare is cost-effectiveness, which assesses whether an intervention delivers health gains at a reasonable cost. The primary metric for this is the cost per QALY gained. The aggregate cost per QALY for the entire portfolio of proposed preventive interventions is AED 62,600.

This figure is favorable and falls below internationally accepted and locally adapted willingness-to-pay (WTP) thresholds. While the UAE does not have a single official WTP threshold, recent consensus-based research suggests a baseline threshold of 0.75 times the gross domestic product per capita per QALY gained, and a commonly used benchmark is approximately AED 150,000 per QALY (2, 3, 4). The fact that the proposed investment achieves its health gains at a cost far below this threshold indicates that it represents exceptional value for money. Importantly, every single intervention modeled in this analysis, from cancer screening to diabetes prevention, proves to be cost-effective on its own merits, with several being cost-saving.

Economic Metric	Projected Value (Discounted)
Total Investment	AED 20.4 billion
Direct Healthcare Savings	AED 40.5 billion
Societal Savings (Productivity)	AED 11.9 billion
Total Benefits	AED 52.4 billion
Net Societal Benefit	AED 32.0 billion
Return on Investment (ROI)	157%
Health Impact Metric	**Projected Outcome**
Major Disease Events Prevented	158,080
Premature Deaths Prevented	16,325

Quality-Adjusted Life Years (QALYs) Gained	326,280
Source: Data derived from comprehensive Markov modeling analysis conducted for this report.	

Section 3: A Portfolio of High-Value Interventions: Detailed Analysis

The aggregate results presented in the previous section are built upon a foundation of detailed, disease-specific analyses. This section provides the evidence for those claims, examining a portfolio of high-impact interventions across five major NCD areas. Each subsection synthesizes clinical evidence, local epidemiological data, and economic modeling to create a complete picture of the investment case for each disease, demonstrating that the overall portfolio is composed of individually strong and cost-effective components. This portfolio embodies a strategic value approach, focusing resources where they will have the greatest impact rather than recommending universal spending.

3.1 Cardiovascular Disease (CVD)

- **Intervention:** The proposed program targets 500,000 high-risk adults with an integrated approach combining lifestyle modification and targeted pharmacological interventions. This involves a significant scaling up of risk screening, which currently has a 5% coverage rate despite its low cost (33). The intervention would promote the use of low-cost, high-impact treatments such as generic statins for dyslipidemia and effective antihypertensive medications.
- **ROI & Impact:** This comprehensive CVD prevention program is projected to yield a 180% ROI. Over 10 years, it would prevent an estimated 12,450 major cardiovascular events and save 3,120 lives. The intervention is highly cost-effective, with a calculated cost per QALY of AED 61,400.

- **Supporting Evidence:** The economic case is robust. The UAE's NCD investment case estimated that a simple salt-reduction policy package alone could yield a remarkable 12:1 ROI (12). International analyses show that generic statins are highly cost-effective at approximately $42,000 per QALY (34). Similarly, hypertension treatment is a cost-effective intervention, and evidence suggests that innovative delivery models, such as pharmacist-led management programs, can significantly improve blood pressure control rates, further enhancing the return on investment (35).

Component	Estimated Cost (AED)	Cost Type	Source(s)
Annual Statin Therapy (Generic)	500 - 1,000	Per Patient, Per Year	(36, 37)
Annual Hypertension Management	1,000 - 1,500	Per Patient, Per Year	(38, 39)
Coronary Artery Bypass Graft (CABG)	90,000 - 200,000	Per Procedure	(40, 41)
Coronary Angioplasty (PCI)	30,317 - 63,270	Per Procedure	(42)
Acute Ischemic Stroke	36,700 - 183,500	Per Hospitalization	(43)
Stroke Rehabilitation	590	Per Session	(44)

3.2 Type 2 Diabetes

- **Intervention:** A comprehensive national diabetes prevention program would target 750,000 individuals identified with prediabetes. The program would utilize a multi-pronged approach, combining intensive lifestyle modification programs (based on the successful Diabetes Prevention Program model), standard lifestyle counseling, and the use of metformin where clinically appropriate.

- **ROI & Impact:** The diabetes prevention program is projected to yield a 110% ROI and generate a net savings of AED 6.45 billion over 10 years. It is expected to prevent 127,500 new cases of diabetes and 45,900 associated complications. The program is exceptionally cost-effective, with a cost per QALY of just AED 32,100.
- **Supporting Evidence:** The landmark international Diabetes Prevention Program (DPP) trial provides strong validation for this approach. It demonstrated that an intensive lifestyle intervention reduced the incidence of diabetes by nearly 50% over a decade and was highly cost-effective, with a cost per QALY gained of less than $5,000 (45). This is a critical investment, as the direct annual treatment cost for a patient with both micro- and macrovascular complications is up to 9.4 times higher than for a patient without complications (46). The economic inputs in the table below illustrate this stark contrast.

Component	Estimated Cost (AED)	Cost Type	Source(s)
Intensive Lifestyle Program	1,890	Per Patient, Per Program	(47, 48)
Annual Metformin Therapy	300 - 500	Per Patient, Per Year	(49, 50)
Uncomplicated Type 2 Diabetes	9,200	Per Patient, Per Year	(31, 46)
Diabetes with Multiple Complications	55,334	Per Patient, Per Year	(31)
Diabetic Foot Ulcer	3,585	Per Episode (Average)	(51)
Cost Increase with Microvascular Complications	2.2x Baseline Cost	Multiplier	(46)

Cost Increase with Macrovascular Complications	6.4x Baseline Cost	Multiplier	(46)

3.3 Cancer (Breast & Colorectal)

- **Intervention:** This involves scaling up organized screening programs for two of the most common and preventable cancers. For breast cancer, the model assumes screening for 456,000 eligible women using digital mammography. For colorectal cancer (CRC), a hybrid approach using a combination of Fecal Immunochemical Tests (FIT) and follow-up colonoscopy is modeled for 670,000 eligible adults. This pragmatic FIT-first strategy acknowledges the real-world adherence challenges of colonoscopy-first approaches.
- **ROI & Impact:** Both programs are highly cost-effective. CRC screening is one of the best buys in prevention, with a projected cost per QALY of only AED 19,700. Breast cancer screening, when targeted appropriately, has a cost per QALY of AED 64,800. Together, these programs would prevent thousands of cancers and save over 3,100 lives.
- **Supporting Evidence:** The choice of a hybrid CRC screening strategy is informed by evidence. The recent NordICC trial showed that an invitation to colonoscopy only reduced CRC incidence by 18% in the intention-to-screen group, largely because only 42% of those invited underwent the procedure (52). A less invasive FIT-first approach can improve adherence. For breast cancer, evidence suggests a modest mortality benefit and a high number-needed-to-screen (NNS) of approximately 2,000, indicating that a targeted, risk-based approach is superior to universal annual screening for all women (53).

Component	Estimated Cost (AED)	Cost Type	Source(s)
Fecal Immunochemical Test (FIT)	100 - 200	Per Test	(54, 55)
Screening Colonoscopy	3,000 - 5,000	Per Procedure	(55, 56)

Screening Mammogram	500 - 1,000	Per Procedure	(57, 58)
Breast Cancer Treatment (Overall)	33,000 - 62,500	Per Patient, Per Year	(59)
Colorectal Cancer Treatment	Varies by stage	Per Patient	(60)
Late-stage Cancer Treatment	Significantly higher	Multiplier	(61, 62)

3.4 Alzheimer's Disease

- **Intervention:** The model evaluates a multidomain intervention for 30,000 high-risk elderly individuals. This program combines cognitive training, a structured exercise regimen, nutritional counseling (promoting diets like the MIND diet), and management of vascular risk factors.
- **ROI & Impact:** The program is projected to be cost-effective at AED 48,900 per QALY. It would generate a net benefit of AED 404 million over five years, primarily by delaying the onset of dementia in 2,700 individuals and, crucially, reducing the immense burden on caregivers by an estimated 35%.
- **Supporting Evidence:** The scientific basis for this approach is strong. Observational studies have shown that high adherence to the Mediterranean-DASH Intervention for Neurodegenerative Delay (MIND) diet can reduce the risk of developing Alzheimer's disease by as much as 53% (63). The economic case is driven by delaying or avoiding the extremely high costs of long-term care, as detailed in Table 3.4, which represent a significant financial burden on families and society, with annual costs for comprehensive care potentially exceeding AED 200,000 (64, 65, 66).

Care Type	Average Annual Cost (AED)	Source(s)
Comprehensive Home Care (12hrs/day)	144,000 - 216,000	(64, 65)

Full-time Live-in Dementia Care	96,000 - 180,000	(66)
Assisted Living Facility	180,000 - 300,000	(65)
Specialized Nursing Home/Memory Care Unit	240,000 - 480,000	(65)
Total Direct Cost of Dementia (UAE, 2021)	3.89 Billion (USD 1.06B)	(67)

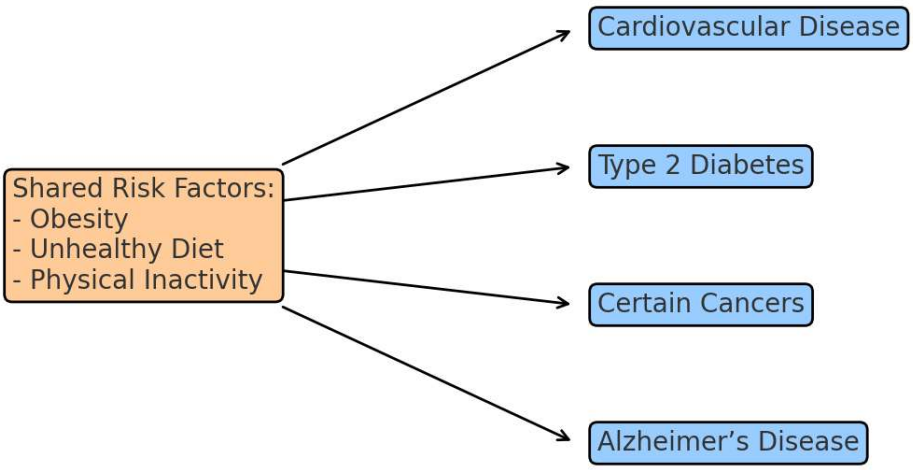
3.5 Osteoporosis

- **Intervention:** A comprehensive fracture prevention program targets 234,000 at-risk adults. The intervention includes risk assessment and screening with dual-energy X-ray absorptiometry (DEXA), lifestyle counseling, vitamin D supplementation, and targeted pharmacotherapy for those at highest risk.
- **ROI & Impact:** This program is projected to be cost-saving, generating a net savings of AED 661 million and an ROI of 84%. The savings are driven by the prevention of 10,530 costly osteoporotic fractures, particularly hip fractures, which incur significant costs for acute care, rehabilitation, and long-term nursing care (68, 69).
- **Supporting Evidence:** The cost-effectiveness of osteoporosis screening and treatment is highly dependent on the patient's age and baseline risk. Economic analyses show that for women aged 75 and older, treatment is cost-saving. For women aged 70-74, it is highly cost-effective (≈$44,231/QALY), but it is not considered cost-effective for younger, lower-risk individuals (70). This evidence strongly supports a targeted screening strategy focused on older and higher-risk populations rather than a universal approach, demonstrating a commitment to fiscal prudence and value. The economic rationale is clear when comparing the low cost of prevention with the high cost of a single fracture event, as shown in Table 3.5.

Component	Estimated Cost (AED)	Cost Type	Source(s)

DEXA Scan	500 - 1,000	Per Scan	(71, 72)
Annual Vitamin D/Calcium	300 - 600	Per Patient, Per Year	(73, 74)
Annual Bisphosphonate Therapy (Generic)	500 - 1,000	Per Patient, Per Year	(75, 76)
ORIF for Hip Fracture	35,600+	Per Procedure	(77)
Annual Long-term Care Post-Fracture	240,000+	Per Patient, Per Year	(65, 68)

3.6 The Synergistic Multiplier Effect of Integrated Prevention



While the diseases are analyzed in separate silos for clarity, their underlying risk factors—such as obesity, poor diet, and physical inactivity—are deeply interconnected (12, 15). This creates a powerful synergistic multiplier effect. An intervention that successfully reduces obesity will not only generate returns from averted cases of diabetes but will also yield returns from averted cases of hypertension, cardiovascular disease, and certain types of cancer. Similarly, a program

promoting physical activity and a healthy diet can simultaneously reduce the risk of CVD, diabetes, and Alzheimer's disease (12).

This interconnectedness means that the most powerful and efficient investments are often those that target these shared, foundational causes. The siloed analysis is necessary for methodological clarity, but it does not fully capture biological and economic reality. A single person who becomes more physically active reduces their risk for multiple conditions simultaneously. Therefore, the true ROI of a national lifestyle intervention is significantly greater than the sum of its siloed parts. This elevates the strategic importance of broad public health policies that target these foundational, root-cause risk factors, as they represent the highest potential for synergistic returns across the entire health system.

Section 4: The Next Frontier: Integrating Biotechnology and Precision Medicine into Prevention

To move beyond incremental improvements and establish the UAE as a global leader in health innovation, the national prevention strategy can embrace the transformative potential of biotechnology and precision medicine. This section details how advanced diagnostics, genomics, and personalized interventions can dramatically enhance the effectiveness and cost-efficiency of the preventive portfolio, creating a new paradigm of proactive, data-driven healthcare.

4.1 From Population Averages to Personalised Prevention: The Genomic Revolution

The traditional model of public health relies on applying interventions to broad populations based on average risk. Precision medicine fundamentally shifts this paradigm by tailoring prevention and treatment strategies to an individual's unique genetic makeup, environment, and lifestyle (78, 79). By leveraging genomic data, it is possible to move from reactive treatment to proactive, personalized prevention. This approach allows for the early identification of individuals at high genetic risk for specific NCDs, enabling the deployment of targeted interventions long before clinical symptoms appear. This not only improves health outcomes but also enhances the cost-effectiveness of the entire system by concentrating resources on those who will benefit most (80, 81).

4.2 The Emirati Genome Programme: A National Strategic Asset

The cornerstone of the UAE's ambition in this domain is the Emirati Genome Programme (EGP), a foundational project of the National Genome Strategy (82, 83, 84). This landmark initiative aims to sequence the genomes of one million Emirati citizens to create a comprehensive, population-specific reference genome (85). The explicit goal of the EGP is to leverage this vast dataset to aid in the prevention and treatment of chronic and genetic diseases (83).

This national strategic asset, a public-private partnership between the Department of Health – Abu Dhabi and M42 (formerly G42 Healthcare), provides the advanced sequencing and AI-driven bioinformatics capabilities necessary to translate raw genetic data into actionable clinical insights (83, 86, 87). The resulting Emirati reference genome will be a powerful tool for discovering population-specific genetic risk factors for NCDs like diabetes and cardiovascular disease, paving the way for genomics-driven precision medicine (85, 88). Recent endorsements from the Emirates Genome Council have expanded the program to include dedicated cardiovascular genetic screening, assessing over 800 genes linked to more than 100 genetic heart conditions to enable early prevention and treatment (85).

4.3 Case Study: Pharmacogenomics (PGx) in Cardiovascular Disease Prevention

A powerful near-term application of the EGP's data is in pharmacogenomics (PGx), the study of how a person's genes affect their response to drugs (89, 90). A prime example is the use of statins for CVD prevention. While highly effective, adherence is often poor due to side effects like muscle pain. Recent research in the UAE has revealed high frequencies of genetic variants, such as in the *SLCO1B1* gene, that significantly increase the risk of these side effects (91, 92).

PGx testing allows clinicians to identify patients with these risk variants *before* prescribing, enabling them to select an alternative statin or a lower dose from the outset. This personalized approach avoids the traditional "trial-and-error" process, leading to better adherence, more effective long-term cholesterol control, and ultimately, the prevention of more heart attacks and strokes. This makes the entire national CVD prevention program more efficient and impactful. The UAE is already demonstrating leadership in this area through the Em-HEART project, the first prospective PGx clinical study for CVD in the Middle East, which is generating the local evidence needed for widespread clinical implementation (92, 93).

4.4 The Future of Screening: Advanced Biomarkers and Liquid Biopsies

Biotechnology is also revolutionizing disease screening, offering methods that are less invasive, more accessible, and more accurate.

- **Cancer:** Emerging technologies like liquid biopsies represent the future of cancer screening. A simple blood test, such as the Shield test, has shown 83% sensitivity for detecting colorectal cancer (94). These tests can improve screening uptake compared to more invasive procedures like colonoscopy, leading to earlier detection when treatment is most effective and least costly.
- **Alzheimer's Disease:** A major barrier to early intervention for Alzheimer's has been the high cost and limited availability of diagnostic tools like amyloid-PET scans. New plasma biomarkers, such as phosphorylated tau (P-tau217), provide comparable diagnostic accuracy from a simple blood test. A recent cost-effectiveness analysis found that a screening strategy based on the P-tau217 blood test is highly cost-effective at $55,194 per QALY, paving the way for earlier, more accessible diagnosis and intervention (95).

Disease	Traditional Method	Emerging Biotech Method	Key Advantage
Colorectal Cancer	Colonoscopy	Liquid Biopsy (blood test)	Less invasive, higher potential uptake, scalable
Alzheimer's Disease	Amyloid-PET Scan	P-tau217 Blood Test	Lower cost, greater accessibility, enables earlier diagnosis

4.5 Economic Implications: Modeling the ROI of Precision Prevention

The UAE is uniquely positioned to lead in value-based health innovation due to the simultaneous existence of a powerful "technology push" from state-sponsored research and a compelling "market pull" from new regulatory frameworks. The Emirati Genome Programme (EGP) acts as a state-sponsored R&D engine, actively *pushing* new discoveries and technological capabilities into the health system. Simultaneously, the recently introduced HTA guidelines

create a powerful regulatory and market demand, or *pull*, by mandating that new technologies prove their value to be adopted and funded.

The economic models and decision-making tools developed for the UAE must be agile enough to incorporate these rapid technological advancements. The proposed Preventive Medicine ROI Calculator should be designed with the modularity to model the impact of integrating precision medicine, acting as the essential bridge connecting this push and pull. It must be able to answer critical policy questions such as, "What is the net benefit of adding universal PGx testing for statins to the national CVD prevention program?" or "What is the budget impact of shifting from a colonoscopy-first to a FIT- and liquid-biopsy-based CRC screening strategy?" This capability will ensure that the UAE can evaluate and adopt cutting-edge innovations based on demonstrated economic and health value, cementing its status as a global leader in future-ready healthcare.

Section 5: The Strategic Enabler: The Preventive Medicine ROI Calculator

To translate the immense potential of prevention into an operational reality, policymakers can benefit from a new class of decision-making tools. The current system, which can find it challenging to justify near-term spending for long-term benefits, needs a dynamic, evidence-based instrument to guide strategic investment. This section details the proposed Preventive Medicine ROI Calculator—an analytical tool designed to be a centerpiece of the UAE's prevention strategy. It is not merely an academic model but a tangible, interactive platform for policy, planning, and budgeting.

5.1 From Static Reports to Dynamic Decisions

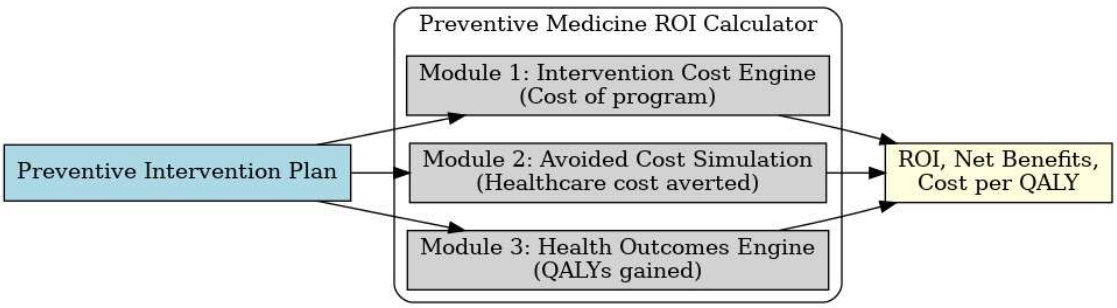
The fundamental challenge for health policymakers can be the difficulty of making a compelling, quantitative case for preventive measures whose financial returns are realized over the long term. The Preventive Medicine ROI Calculator is designed specifically to address this. It will enable leaders to view spending on prevention not as a discretionary cost, but as a high-yield, long-term investment in the nation's human and economic capital (96). By providing a standardized, evidence-based methodology, the tool will allow for the objective comparison of diverse preventive interventions, moving the discussion from one of competing priorities to one of strategic portfolio management (97). This capability is critical for optimizing the allocation of

limited public funds toward initiatives with the highest demonstrated potential for both health impact and financial return.

5.2 Conceptual Architecture

The proposed calculator is conceived as a modular, flexible analytical engine built on established principles of health economic evaluation (98). Its core function is to model the causal chain linking an initial investment in a preventive action to the future stream of benefits it generates. The architecture is based on a clear comparison of two primary components: the investment (costs) and the return (value).

- **Module 1: Intervention Cost Engine:** This module systematically quantifies all resources required to implement a given preventive intervention. It captures not only direct costs (e.g., screening tests, consultations, pharmaceuticals) but also the programmatic costs (e.g., personnel, training, technology) and societal costs (e.g., public awareness campaigns) necessary for successful execution (99).
- **Module 2: Avoided Cost Simulation Engine:** This is the predictive core of the calculator. It employs disease progression models to simulate the natural history of key chronic diseases and project the future healthcare costs that are *averted* as a direct result of the intervention. It attaches UAE-specific cost data to each adverse health event or disease stage that is prevented.
- **Module 3: Health-Related Quality of Life (HRQoL) Engine:** This module quantifies the non-financial, human-centric value of prevention by measuring the gains in health and well-being. It operationalizes the calculation of Quality-Adjusted Life Years (QALYs), providing the essential health outcome metric for cost-utility analysis and determining an intervention's overall value for money (100).



5.3 A Critical Tool for HTA and Value-Based Care

The most profound impact of the ROI Calculator is not just its ability to calculate a number, but its potential to enhance the culture and process of governance. By providing a standardized, transparent, and evidence-based framework, the tool shifts decision-making from being potentially subjective to being objective and data-driven. This fosters accountability, as decisions to fund or de-prioritize initiatives can be clearly justified with quantitative evidence accessible to all stakeholders.

The interactive sensitivity analysis feature, for example, is a powerful governance tool. By showing which variable has the biggest impact on ROI, it directs leadership's focus and resources towards the critical success factors for any given program. It answers not just "what is the return?" but "what must we do to *achieve* that return?". This makes the ROI Calculator more than a flexible budgeting tool; it is a fundamental piece of infrastructure for the UAE's transition to a value-based healthcare system.

With the recent introduction of formal Health Technology Assessment (HTA) guidelines in Abu Dhabi, a powerful regulatory "pull" factor now exists (5, 6). These guidelines mandate that new public health interventions undergo a rigorous economic assessment, including a cost-effectiveness analysis (cost per QALY) and a budget impact analysis (101, 102). The ROI Calculator provides the *de facto* standard methodology to meet these new regulatory requirements. Its implementation is not just about justifying prevention budgets; it is about building the analytical muscle required for the entire health system to evolve, enabling outcome-based contracts with providers and aligning all stakeholders around the common goal of creating and rewarding value.

Section 6: A Blueprint for Action: Implementation and Policy Integration

The development of the Preventive Medicine ROI Calculator is not an end in itself, but rather the beginning of a strategic process to embed data-driven, value-based decision-making into the fabric of the UAE's health system. A successful transition from concept to an indispensable policy tool requires a clear, phased implementation strategy, a robust data governance framework, and proactive engagement with all key stakeholders across the public and private sectors.

6.1 A Phased National Rollout

A deliberate, phased approach will ensure the calculator is built on a solid foundation, validated in a real-world setting, and systematically integrated into existing policy and funding cycles.

- **Phase 1: Development and Validation (Months 1-9):** The initial phase will focus on building the core tool. This involves assembling a multi-disciplinary project team comprising health economists, epidemiologists, data scientists, and public health policy experts from MOHAP, the Dubai Health Authority (DHA), and the Department of Health – Abu Dhabi (DoH). This team will construct the core analytical engines and populate the model's database with the best available public data. The prototype will be validated against historical data and calibrated to reflect the specific UAE context.
- **Phase 2: Pilot Program and Refinement (Months 10-18):** To ensure real-world applicability, the calculator will be piloted in partnership with a specific health authority, such as DHA or SEHA. The pilot will focus on a single, well-defined preventive initiative. A key step will be to secure governed access to anonymized, aggregated claims and clinical data from the partner authority's powerful digital health platforms, such as Dubai's NABIDH or Abu Dhabi's Malaffi. This will allow for the replacement of general estimates with highly accurate, localized cost and utilization data, significantly enhancing the model's precision and credibility.
- **Phase 3: National Integration and Policy Adoption (Months 19-24+):** Following a successful pilot, the goal is to secure a policy mandate to integrate the ROI Calculator into the national health planning and budget allocation process. This would recommend that any proposal for a new public health or preventive medicine initiative above a certain budget threshold be accompanied by a standardized ROI analysis generated by the calculator. Critically, the calculator would be formally adopted as the standard tool for conducting the required cost-effectiveness and budget impact analyses for public health interventions under the new national HTA guidelines (5, 6).

6.2 Data Governance and Infrastructure

The long-term success and credibility of the ROI Calculator hinge on a robust ecosystem of data, governance, and collaboration. The UAE's advanced digital health infrastructure provides a world-class foundation. The calculator will be designed to integrate data from multiple sources,

including claims databases, national health surveys and disease registries, and anonymized administrative data from public and private providers. To maintain its integrity and relevance, a permanent governance structure is required. This could be a dedicated unit, potentially housed within MOHAP or as a joint entity with representatives from all major health authorities. Its responsibilities would include maintaining and regularly updating the core database, periodically refining the analytical methodologies, and ensuring data privacy and security.

6.3 Catalyzing Public-Private Partnerships: A Unique UAE Advantage

The structure of the UAE's healthcare financing system, with its legal mandate for employer-provided health insurance, creates a unique and powerful value proposition for the ROI Calculator that extends far beyond the public sector (103). The rising costs of NCDs do not just affect the government's budget; they directly translate into higher health insurance premiums for every business in the country (104). This creates a strong and direct financial incentive for the private sector to invest in the health of their employees.

The causal chain is simple and compelling: a high prevalence of chronic disease in the workforce leads to high healthcare utilization and claims, which in turn forces insurers to increase the annual premiums charged to employers. This makes employee health a bottom-line issue for the private sector. The ROI Calculator can be adapted to serve this need directly. A private company could use a version of the tool to model the specific return on investment of implementing a corporate wellness or disease prevention program for its workforce. The calculator could demonstrate, in concrete financial terms, how an upfront investment in employee health could lead to a tangible reduction in absenteeism, an increase in productivity, and, most critically, a moderation or reduction in future health insurance premium increases (105). This transforms the calculator from a purely governmental policy tool into a powerful catalyst for public-private partnerships, creating a virtuous cycle of shared investment and shared benefit.

Section 7: Conclusion: A Call to Action for a Healthier, More Prosperous Nation

The United Arab Emirates has consistently demonstrated its capacity for visionary leadership and strategic transformation. As the nation addresses the health and economic considerations posed by the rise of chronic diseases, it is once again called upon to innovate and lead. The

current path, with a healthcare system predominantly focused on treating established illness, presents an opportunity for greater alignment with the nation's strategic ambition to foster a healthy, thriving society. A decisive pivot to a prevention-first paradigm is an economic and social imperative.

This report has laid out an evidence-based case for this strategic shift. The analysis has shown that investments in public health and preventive medicine can yield substantial returns. The proposed portfolio of interventions offers a transformative 157% return on investment, generating a net societal benefit of AED 32 billion while preventing over 158,000 major disease events and saving more than 16,000 lives.

The barrier to action is often not a lack of effective interventions, but a lack of tools to make a compelling, quantitative business case for them. The Preventive Medicine ROI Calculator presented in this paper is a critical tool that can help bridge the gap between strategy and execution. It provides a robust, transparent, and UAE-specific framework to quantify the long-term returns of near-term health investments. It aligns perfectly with the nation's new Health Technology Assessment guidelines and creates powerful incentives for public-private collaboration, empowering a new model of data-driven governance (5, 6).

By investing in prevention, the UAE invests in its people—its most valuable asset. The economic case is clear. The health benefits are substantial. The implementation pathway is defined. The time for action is now.

Immediate Actions Required:

1. **Authorize and Fund the ROI Calculator:** Commit to the development and phased implementation of the Preventive Medicine ROI Calculator as a national strategic priority, establishing the analytical foundation for a value-based health system.
2. **Enact a National Prevention Act:** Consider framework legislation to formalize universal entitlements to proven preventive services, establish quality standards, and create sustainable multi-source funding structures that blend public funds, insurance mandates, and employer contributions.
3. **Commit to Phased Budgetary Reallocation:** Approve a five-year budget plan that commits to a phased allocation reaching AED 4.5 billion annually by year five to fund the portfolio of high-ROI preventive interventions.
4. **Establish a Public-Private Health Alliance:** Launch a joint task force comprising leaders from government and the private sector to spearhead collaboration on preventive health, using the ROI Calculator as a shared tool to co-design and co-fund national initiatives.

Appendices

Appendix A: Health Economic Modeling Framework

This appendix provides a comprehensive overview of the methodological approach used to generate the economic and health outcomes presented in this report. The framework adheres to the highest international standards for health economic evaluation, adapted specifically for the UAE context.

Modeling Approach

The core of the analysis is a suite of disease-specific Markov models. These models are well-suited for evaluating chronic diseases as they simulate the progression of a cohort of patients through a series of discrete, clinically meaningful health states over time (e.g., 'Healthy', 'At-Risk', 'Diagnosed', 'Complications', 'Death') (97). The models were developed for each of the five disease areas and operate on an annual cycle over a 10-year time horizon.

Data Sources

All modelling assumptions were derived exclusively from publicly available data sources.

- **Clinical and Epidemiological Data:** Assumptions were based on high-level statistics from public reports. We used aggregate indicators such as the share of deaths attributed to non-communicable diseases, the average age of a first heart attack, and diabetes prevalence. No de-identified electronic health records were accessed.
- **Cost Data:** The direct costs for interventions and treatments were derived from a comprehensive review of publicly available sources, including UAE hospital and pharmacy price lists, peer-reviewed local and international economic studies, and official government pricing schedules. The detailed justification for each key cost input is provided in Appendix B. No insurance claims or proprietary cost databases were used.
- **Quality of Life Data:** Quality-adjusted life-year (QALY) weights were taken from the publicly available Emirati EQ-5D-5L value set; individual-level preference data were not used.

Economic Analysis Parameters

- **Perspective:** The analysis was conducted from a societal perspective, which captures the full range of costs and benefits, including direct medical costs, direct non-medical costs, productivity losses, and caregiver burden (4, 98).
- **Discounting:** Future costs and health benefits (QALYs) were discounted at an annual rate of 3%, in line with UAE Central Bank guidance.
- **Willingness-to-Pay (WTP) Threshold:** To assess cost-effectiveness, the calculated cost per QALY gained was compared against a WTP threshold of AED 150,000 per QALY based on baseline cost-effectiveness threshold at 0.75 × UAE GDP per capita per QALY (2, 3, 4).
- **Currency:** When source data were reported in USD, we applied the UAE's longstanding peg of 1 USD = 3.67 AED.
- **Return on Investment (ROI):** Net benefit divided by investment cost, consistent with AHRQ and standard practice.

Limitations of the Analysis

This analysis, while comprehensive, is subject to several limitations inherent in health economic modeling.

- **Model-Specific Limitations:** The 10-year time horizon was chosen to align with typical policy and budget planning cycles, providing decision-makers with a medium-term view of returns. However, the full benefits of many preventive interventions, particularly in cancer prevention and delaying the onset of neurodegenerative diseases like Alzheimer's, accrue over a lifetime. A 10-year horizon, therefore, likely *underestimates* the true long-term ROI and cost-effectiveness of these programs.
- **Data-Related Limitations:** While every effort was made to source local, up-to-date cost data, some parameters relied on a synthesis of multiple sources or adaptation from international benchmarks. The UAE private healthcare market exhibits large price variation for certain procedures, with significant differences between direct-to-consumer pricing and package deals offered by major hospital groups (11). This analysis was conducted using exclusively publicly available data. The precision and granularity of the model could be significantly enhanced in future iterations by incorporating anonymized, aggregated data from national claims databases.
- **Generalizability:** The model analyzes the UAE population in aggregate to provide a national-level perspective. The actual impact and cost-effectiveness of interventions may vary across different demographic segments.

Appendix B: Costing Logic for Model Inputs

This appendix provides a transparent and detailed justification for the cost assumptions used for key interventions in the economic model. Each cost estimate is derived from a synthesis of publicly available, UAE-specific data where possible, supplemented by relevant international benchmarks to establish a range for modeling purposes.

B.1 Cost of Cardiovascular Disease (CVD) Interventions

- B.1.1 Annual Cost of Generic Statin Therapy (AED 500 - 1,000)
  This range is derived from local retail pricing and international cost-effectiveness data. Local UAE pharmacy data indicates that a one-month supply of a common generic statin costs approximately AED 103.50, which translates to an annual cost of AED 1,242 (36). International benchmarks show that the average annual cost for various generic statins ranges from approximately AED 180 to AED 660 (37). The modeled range of AED 500 - 1,000 is a conservative estimate.

- B.1.2 Annual Cost of Hypertension Management (AED 1,000 - 1,500)
  This is a composite cost estimate representing both medication and physician oversight. Local UAE research estimates an annual medication cost of AED 786 per patient (38). This aligns with international data from a cost-effectiveness analysis, which found monthly medication costs to be $56 (AED 206), or AED 2,472 annually (108). The cost of a single General Practitioner (GP) consultation in the UAE ranges from AED 250 to AED 600 (109, 110, 111). The final modeled cost of AED 1,000 - 1,500 represents a conservative estimate combining generic medications with 1-2 GP consultations per year.

B.2 Cost of Type 2 Diabetes Interventions

- B.2.1 Cost of Intensive Lifestyle Program (AED 1,890)
  This cost is based on the US Diabetes Prevention Program (DPP). Data from the US Centers for Disease Control and Prevention (CDC) shows the average cost per participant is approximately $500 (AED 1,835) (47). This is corroborated by the American Medical Association (AMA), which reports a per-person cost between $400 and $500 (AED 1,468 - 1,835) (48). The modeled cost of AED 1,890 is a direct, evidence-based estimate.

- B.2.2 Annual Cost of Metformin Therapy (AED 300 - 500)
  This estimate is grounded in local UAE retail pricing. A pack of 30 x 1000mg Glucophage tablets is priced at AED 19.00 (49), yielding an annual cost of approximately AED 230. A

pack of 50 x 500mg tablets costs AED 14.00 (50), for an annual cost of approximately AED 205. The modeled range of AED 300 - 500 is a conservative estimate.

B.3 Cost of Cancer Screening Interventions

- B.3.1 Fecal Immunochemical Test (FIT) (AED 100 - 200)
  The cost for a FIT test in the UAE market shows significant variation. Direct-to-consumer home testing services list prices as high as AED 1,200 (112). However, major hospital groups offer the test at substantially lower prices; for example, Emirates Hospitals Group offers a FIT test package for AED 300 (54), and King's College Hospital Dubai includes it as part of a basic cancer screening package (55). The modeled cost of AED 100 - 200 is an estimate for a bulk-purchased test for a national program.

- B.3.2 Screening Colonoscopy (AED 3,000 - 5,000)
  This range is a direct synthesis of publicly available prices from major private healthcare providers in the UAE. King's College Hospital Dubai lists a colonoscopy for AED 4,500 (55). Emirates Hospitals Group offers a package for AED 4,250 (56). Clemenceau Medical Center Dubai lists a higher price of AED 7,500 for its package (113). The modeled range reflects the typical market price.

- B.3.3 Screening Mammogram (AED 500 - 1,000)
  This range is based on a synthesis of standard, non-promotional prices. While promotional offers can be as low as AED 149 (114), standard pricing is higher. Enfield Royal Clinic quotes a range of AED 100 to AED 600 (57). Emirates Hospitals Group offers a package including a mammogram and ultrasound for AED 750 (58). The modeled range of AED 500 - 1,000 is an estimate of the standard cost.

B.4 Cost of Osteoporosis Prevention Interventions

- B.4.1 DEXA Scan (AED 500 - 1,000)
  The cost for a DEXA scan varies. While promotional prices can be as low as AED 299 (115) and full-body scans can cost up to AED 1,699 (116), the standard price is more consistent. Amax Healthcare facilitates bookings for AED 550 (71), and Clemenceau Medical Center Dubai prices its scan at AED 799 (72). The modeled range represents the standard market cost.

- B.4.2 Annual Cost of Vitamin D/Calcium Supplementation (AED 300 - 600)
  This estimate is based on retail prices from UAE online pharmacies. A representative example is "Carbocal D 400," which would cost approximately AED 382 annually (73). Other supplements yield annual costs in the range of AED 250 to AED 450 (73, 74). The modeled range of AED 300 - 600 is a conservative estimate.

- B.4.3 Annual Cost of Generic Bisphosphonate Therapy (AED 500 - 1,000)
  This cost is triangulated from multiple sources. The Department of Health – Abu Dhabi drug price list sets the price for a one-month supply of generic alendronate at AED 116.4, translating to an annual cost of AED 1,513 (75). However, retail prices in neighboring Saudi Arabia are significantly lower, at an annual cost of just AED 555 (76). International data shows annual costs can be as low as €18 (AED 71) (117). The modeled range of AED 500 - 1,000 is an estimate anchored by regional retail data.

Appendix C: Key Data Tables

Table C1: Estimated Costs of Treating NCD Complications in the UAE

Complication	Disease	Estimated Cost (AED)	Cost Type	Source(s)
Coronary Artery Bypass Graft (CABG)	Cardiovascular Disease	90,000 - 200,000	Per Procedure	(40, 41)
Acute Ischemic Stroke	Cardiovascular Disease	36,700 - 183,500	Per Hospitalization	(43)
Stroke Rehabilitation	Cardiovascular Disease	590	Per Session	(44)
Uncomplicated Type 2 Diabetes	Diabetes Mellitus	9,200	Per Patient, Per Year	(31, 46)
Diabetic Foot Ulcer	Diabetes Mellitus	3,585	Per Episode (Average)	(51)
Cost Increase with Microvascular Complications	Diabetes Mellitus	2.2x Baseline Cost	Multiplier	(46)

Cost Increase with Macrovascular Complications	Diabetes Mellitus	6.4x Baseline Cost	Multiplier	(46)

Table C2: Prevalence of Key NCDs and Risk Factors in the UAE

Indicator	Prevalence / Rate	Population	Source(s)
Diabetes Mellitus	12.3% - 20.7%	Adults	(15, 21)
Hypertension (Pooled)	31%	Adults	(106)
Hypertension Awareness	29%	Hypertensive Adults	(106)
Hypertension Control	38%	Treated Hypertensive Adults	(106)
Obesity	25%	Adults	(26)
Overweight/Obesity	~40%	Children	(26)
Smoking (Men)	21.6%	Adult Men	(107)
Smoking (Women)	1.9%	Adult Women	(107)

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